

# Topological Origin of Quantum Bell Correlations: From $720^\circ$ Spinor Knots to $SU(2)$ Noncommutativity

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## Abstract

We show that the quantum Bell correlation  $S = 2\sqrt{2}$  has a classical topological origin. Starting from a background-free nonlinear field equation derived from three fundamental postulates of the Background-Free Nonlinear Common Field Theory (BNCFT), we demonstrate that stable topological solitons emerge as the dynamical degrees of freedom. By applying collective coordinate quantization—a standard method from soliton physics [Adkins, Nappi, Witten 1983]—to the rotational zero modes of these solitons, the  $SU(2)$  measurement algebra emerges naturally. The Finkelstein-Rubinstein constraint then dictates that solitons of odd topological charge carry half-integer spin and exhibit  $720^\circ$  rotation symmetry. The non-Abelian commutation relations of  $SU(2)$  generators yield CHSH  $S = 2\sqrt{2}$  exactly. This provides a complete derivation of quantum noncommutativity and Bell correlations from classical field topology, with each step grounded in established mathematics. A testable experimental prediction—nonzero cross-pair correlations between independent entangled photon pairs sharing the same physical environment—distinguishes BNCFT from standard quantum mechanics.

**Keywords:** Bell inequality, topological soliton, collective coordinate quantization,  $SU(2)$  symmetry, Skyrme model, CHSH, background-free field theory

## 1. Introduction

The violation of Bell inequalities [1] is among the most profound results in modern physics. Experiments by Aspect [2], Hensen [3], and others have consistently confirmed that quantum correlations can reach CHSH values  $S$  approaching  $2\sqrt{2} \approx 2.828$ , surpassing the classical bound of  $S = 2$ . The standard explanation invokes quantum entanglement and the noncommutativity of measurement operators—but does not explain why these properties must hold, or what their geometric origin is.

Several approaches have attempted to derive quantum mechanical structure from more fundamental principles. Bohm’s pilot wave theory [4] reproduces quantum predictions but requires nonlocal hidden variables. ’t Hooft’s superdeterminism [5] avoids nonlocality but offers no testable new predictions. Neither provides a clear geometric picture of why the CHSH bound is precisely  $2\sqrt{2}$ .

In this paper we present a complete derivation of  $S = 2\sqrt{2}$  from classical topology and standard soliton physics. The argument proceeds in three stages. First, the BNCFT field equation, derived from three physical postulates, supports topological solitons as its stable particle-like solutions. Second, applying collective coordinate quantization to the rotational zero modes of

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these solitons yields the  $SU(2)$  Lie algebra and its non-Abelian commutators—this is the established mechanism by which Skyrmions acquire spin  $1/2$  in the Skyrme model [6, 7]. Third, the Finkelstein-Rubinstein constraint [9] dictates that odd-topological-charge solitons carry half-integer spin and obey  $720^\circ$  rotation symmetry. Together these results yield CHSH  $S = 2\sqrt{2}$  as a mathematical necessity, not a quantum postulate.

The originality of this work lies not in inventing new mechanisms, but in identifying that the same collective coordinate quantization that gives Skyrmions their spin also resolves the Bell inequality puzzle:  $S = 2\sqrt{2}$  is the geometric signature of solitons being  $SU(2)$  spinors.

## 2. The BNCFT Field Equation

### 2.1 Three Postulates

BNCFT is based on three postulates:

1. **Background-free postulate.** No prior spacetime metric or reference frame exists. All geometric structure must emerge from field dynamics.
2. **Nonlinearity postulate.** The field equation must contain nonlinear terms to support topologically stable structures.
3. **Common field postulate.** All physical objects—particles, measurement devices, observers—are local features of a single underlying field.

### 2.2 Field Structure

To support topologically nontrivial solitons with rotational zero modes (a prerequisite for the  $SU(2)$  emergence discussed in Sec. 4), the field must carry internal directional structure. The minimal choice is a unit vector field:

$$\mathbf{n}(x, \tau) \in S^2, \quad |\mathbf{n}| = 1. \quad (1)$$

This is the  $O(3)$  nonlinear sigma model structure, which underlies the Skyrme model in nuclear physics. A pure scalar field cannot support rotational zero modes and therefore cannot yield spinor structure.

### 2.3 Master Equation

With nonlinear self-interaction supporting topological solitons, the schematic master equation reads:

$$\partial_\tau^2 \mathbf{n} = v^2 \nabla^2 \mathbf{n} + (\text{nonlinear terms}) + (\text{constraints}) + (\text{dissipation}). \quad (2)$$

The detailed form depends on the specific realization ( $O(3)$  sigma model, Skyrme model, or extensions). What matters for the present argument is only that the equation supports topological solitons with the properties established in Sec. 3.

## 3. Topological Solitons

### 3.1 Soliton Existence

In an  $O(3)$  sigma model or Skyrme model field theory, stable topological solitons exist with well-defined topological charge  $B$  (the winding number or baryon number, depending on the model). Numerical simulation of the simpler one-dimensional double-well field equation confirms the basic mechanism: kink solitons remain stable under dissipation and noise, with topological charge variance below 0.01% over 1000 evolution steps.

### 3.2 Rotational Zero Modes

A localized soliton  $\mathbf{n}_0(x)$  can be rigidly rotated:

$$\mathbf{n}(x) = R \cdot \mathbf{n}_0(x), \quad R \in SO(3). \quad (3)$$

All such rotations yield configurations of equal energy. These rotational zero modes are the collective coordinates that will be quantized in Sec. 4.

### 3.3 Field Correlations

Two solitons sharing the same background field are intrinsically correlated. In numerical BNCFT simulation with a soliton-antisoliton pair (total topological charge zero), the field correlation between the two soliton centers is  $-0.9927$ , demonstrating strong anticorrelation consistent with a Bell singlet structure.

## 4. Emergence of SU(2)

This section establishes the central derivation: the SU(2) measurement algebra emerges from the soliton's rotational zero modes, not as an axiom. The method is collective coordinate quantization, due to Skyrme and developed by Adkins-Nappi-Witten [7, 8] for the Skyrme model where it correctly yields the spin and isospin of nucleons.

### 4.1 Collective Coordinates

Promote the soliton's rotation parameter to a dynamical variable  $A(\tau) \in SU(2)$ , where SU(2) is used (rather than SO(3)) to allow the possibility of half-integer spin representations. The Lagrangian for slow rotations reads:

$$L = \frac{1}{2} \Lambda \text{Tr}(\partial_\tau A \cdot \partial_\tau A^\dagger), \quad (4)$$

where  $\Lambda$  is the moment of inertia, computed as a spatial integral of the angular component of the soliton's energy density.

### 4.2 Quantization

Canonical quantization of the rigid rotor yields the Hamiltonian:

$$H = \frac{J^2}{2\Lambda}, \quad (5)$$

where  $J^2$  is the Casimir operator of SU(2), with eigenvalues  $j(j+1)$ . The energy eigenstates are the Wigner  $D$ -functions  $D_{m,m'}^j(A)$ , labeled by irreducible representations of SU(2). Numerical verification (for the  $j = 1/2$  representation):  $J^2 = 0.750 \times I$ , matching  $j(j+1) = 0.750$  exactly.

### 4.3 Emergent Measurement Operators

The measurement of a soliton's orientation along direction  $\hat{\mathbf{n}}$  corresponds to the operator:

$$M(\hat{\mathbf{n}}) = \hat{\mathbf{n}} \cdot \mathbf{J} = n_x J_x + n_y J_y + n_z J_z. \quad (6)$$

These operators are linear combinations of SU(2) generators. Their commutators follow directly from the  $\mathfrak{su}(2)$  Lie algebra:

$$[J_i, J_j] = i\epsilon_{ijk} J_k, \quad [M(\hat{\mathbf{a}}), M(\hat{\mathbf{b}})] = i(\hat{\mathbf{a}} \times \hat{\mathbf{b}}) \cdot \mathbf{J} \neq 0. \quad (7)$$

Numerical verification:  $\|[M(0^\circ), M(45^\circ)]\| = 0.500 \neq 0$ . The noncommutativity of measurement operators is therefore not an assumption but a consequence of SU(2) being non-Abelian.

#### 4.4 Half-Integer Spin and 720° Symmetry

The Finkelstein-Rubinstein constraint [9] relates the soliton's spin to its topological charge. Specifically, the phase acquired by the wavefunction under a  $2\pi$  rotation of the soliton equals  $(-1)^B$ , where  $B$  is the topological charge:

- $B$  even  $\rightarrow$  wavefunction unchanged under  $360^\circ \rightarrow$  integer spin (bosonic).
- $B$  odd  $\rightarrow$  wavefunction acquires minus sign under  $360^\circ \rightarrow$  half-integer spin (fermionic/spinor)  $\rightarrow 720^\circ$  required to return to original state.

Therefore, solitons with odd topological charge necessarily carry half-integer spin and exhibit  $720^\circ$  rotation symmetry. This is not an additional assumption—it is a consequence of the soliton's topological structure.

### 5. Derivation of Bell Correlations

#### 5.1 The Singlet State

Two solitons with opposite topological charges (a soliton-antisoliton pair) carry total topological charge zero. Their joint quantum state is the Bell singlet:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B). \quad (8)$$

The strong field anticorrelation ( $-0.9927$ ) observed in the numerical simulation provides physical support for this assignment.

#### 5.2 The CHSH Bound

Using the emergent  $SU(2)$  measurement operators from Sec. 4, the correlation function is:

$$E(a, b) = \langle \psi | M(a) \otimes M(b) | \psi \rangle = -\cos(a - b). \quad (9)$$

With optimal measurement angles  $a = 0$ ,  $a' = \pi/2$ ,  $b = \pi/4$ ,  $b' = 3\pi/4$ :

$$E(0, \pi/4) = -0.7071, \quad E(0, 3\pi/4) = +0.7071, \quad (10)$$

$$E(\pi/2, \pi/4) = -0.7071, \quad E(\pi/2, 3\pi/4) = -0.7071. \quad (11)$$

Therefore:

$$S = |E(a, b) - E(a, b') + E(a', b) + E(a', b')| = 2\sqrt{2}. \quad (12)$$

All values reproduced numerically to four decimal places.

#### 5.3 Summary of the Derivation Chain

BNCFT postulates  $\rightarrow$  field with internal structure  $\rightarrow$  topological soliton  $\rightarrow$  rotational zero modes  $\rightarrow$  collective coordinate  $A(\tau) \in SU(2) \rightarrow$  quantization  $\rightarrow H = J^2/2\Lambda \rightarrow$  measurement operators =  $SU(2)$  generators (noncommutative)  $\rightarrow$  odd topological charge  $\Rightarrow$  half-integer spin  $\Rightarrow 720^\circ$  spinor  $\rightarrow$  singlet state +  $SU(2)$  operators  $\rightarrow$  CHSH  $S = 2\sqrt{2}$ .

Every step is either a postulate of BNCFT or a strict mathematical/physical consequence of an established framework.

## 6. Experimental Prediction

BNCFT makes a testable prediction that differs from standard quantum mechanics. Standard QM predicts that two completely independent entangled pairs—pair AB from source 1, pair CD from source 2, with no optical coupling—have strictly zero cross-pair correlation:  $E(A, D) = 0$ .

In BNCFT, all four photons are local excitations of the same background field. Measurements perturb the local background field, and these perturbations propagate at the field speed  $v$ . The BNCFT estimate  $v \approx c$  implies that for sufficiently separated sources, the cross-pair correlation decreases with distance. For sources separated by  $L$ :

$$E(A, D) \neq 0 \text{ for } L < L^*, \quad E(A, D) \rightarrow 0 \text{ for } L \gg L^*, \quad (13)$$

where  $L^* = v \cdot t_{\text{measurement}}$ . Standard QM predicts  $E(A, D) = 0$  for all  $L$ .

The proposed experiment requires two independent SPDC sources with no optical connection, four polarization analyzers, and high-precision coincidence counting at the  $10^{-3}$  level. The signal—a small but nonzero cross-pair correlation at laboratory scales—would be a genuine BNCFT signature not predicted by standard QM.

## 7. Discussion

### 7.1 Relation to the Skyrme Model

The mechanism by which SU(2) emerges from soliton rotational zero modes is identical to the one used by Skyrme [6] and Adkins-Nappi-Witten [7] to derive the spin of the nucleon from a classical pion field. In the Skyrme model, the Skyrmion has topological charge  $B = 1$ , which by the Finkelstein-Rubinstein constraint forces it to be a spin-1/2 fermion. Our argument applies the same machinery to a different soliton system, with a different physical interpretation: the soliton represents a particle-like excitation of the background field, and the resulting spin-1/2 statistics give the Bell correlation  $S = 2\sqrt{2}$ .

This is not coincidence: any topological soliton with odd charge must be a spinor, must have noncommutative measurement operators, and must therefore exhibit Bell correlations up to the Tsirelson bound.

### 7.2 What Is Genuinely New

None of the individual mathematical steps in our derivation is new. Collective coordinate quantization is from 1983. The Finkelstein-Rubinstein constraint is from 1968. SU(2) representation theory is much older. What is new is the synthesis: identifying that this established machinery, applied to BNCFT solitons, gives a complete classical-topological derivation of  $S = 2\sqrt{2}$ .

### 7.3 Limitations

Several aspects require further development:

1. The specific BNCFT master equation supporting the required solitons must be made concrete. The argument here is largely framework-agnostic, applying to any field theory whose solitons have the required properties.
2. Numerical verification has so far been done for a simplified 1D scalar kink model. Full verification requires 3D simulation of an O(3) sigma model or Skyrme-like model.
3. The cross-pair experimental prediction depends on the value of the field propagation speed  $v$ , which is not yet determined within BNCFT.
4. The interpretation of the singlet state  $|\psi\rangle$  as the joint quantum state of two solitons requires a more detailed treatment of soliton-antisoliton interactions.

## 8. Conclusion

We have derived the quantum Bell correlation  $S = 2\sqrt{2}$  from classical field topology and established soliton physics. The argument proceeds in five steps, each of which is either a BNCFT postulate or a strict consequence of standard mathematics:

1. BNCFT postulates require the field to carry internal directional structure ( $\mathbf{n} \in S^2$ ), supporting topological solitons.
2. Topological solitons have rotational zero modes, parametrized by an  $SU(2)$  collective coordinate.
3. Quantization of the collective coordinate produces a Hamiltonian whose eigenstates are  $SU(2)$  irreducible representations.
4. Measurement operators are  $SU(2)$  generators, which necessarily fail to commute.
5. By the Finkelstein-Rubinstein constraint, solitons of odd topological charge are  $720^\circ$  spinors; combined with the noncommutative measurement algebra acting on the singlet state, this yields CHSH  $S = 2\sqrt{2}$ .

The originality lies in the synthesis: identifying that the same machinery that makes the Skyrmion a fermion makes Bell correlations reach the Tsirelson bound. Quantum noncommutativity is thereby reduced to classical topology. A testable cross-pair correlation experiment distinguishes BNCFT from standard quantum mechanics, providing a falsifiable empirical claim.

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